Did Generative AI Homogenize Online Writing?

A Multi-Corpus, Topic-Controlled Analysis (2009–2026)

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Abstract

We test the widely held intuition that generative AI (ChatGPT, publicly released in 2022Q4) has made online writing more homogeneous. Using per-quarter diversity and similarity metrics on 30 dated text corpora—26 Stack Exchange communities spanning high- and low-cognitive-load topics, plus Hacker News comments, PubMed oncology abstracts, and arXiv computer-science and mathematics abstracts—and a layered battery of controls (length control, four independent topic controls, a known-AI anchor, and interrupted-time-series significance testing), we find that the dramatic surface-level convergence is a length artifact; that a smaller but statistically real rise in contextual-embedding similarity survives length control in most corpora; and that this residual rise is largely a shift in *what* people write about (topic composition), not a convergence of *how* they write within a topic. We therefore find no evidence of broad stylistic homogenization, but clear evidence of topic-level homogenization: a narrowing of the topic mix and the emergence of a shared, AI-shaped set of themes. We argue the topic-level effect is the more consequential one for epistemic diversity, and introduce a live, open tracker to monitor it going forward.

1 Motivation

Since late 2022, a common complaint has surfaced: *everything online is starting to sound the same.* The em-dashes, the "it's not just X—it's Y," the tidy three-part lists. The worry is that as hundreds of millions of people route their writing through a handful of large language models, the diversity of human expression—and, more importantly, the diversity of *ideas*—quietly collapses toward a shared machine mean.

This matters beyond aesthetics. Three stakes in particular:

- **Epistemic diversity.** If the range of answers to a question narrows, so does the pool of ideas a reader is exposed to. Homogenization of *content* (not just style) is a narrowing of what gets said at all.
- **The integrity of the public record.** Scientific abstracts, legal opinions, and reference sites are the durable, searchable memory of a field. If that record converges toward a model's priors, we lose signal about what humans actually discovered or argued.
- **Search and discovery.** Search engines rank a top-N of results. If those results increasingly repeat one another, the marginal information a searcher gains from reading more of them falls—the web gets wider but shallower.

There is also a prior reason to expect an effect. A meta-analysis of human–AI co-creation studies finds a small but significant homogenization effect that persists after AI use ends [1]. And from behavioural economics, when people face more options or hold weaker preferences they default to

the status quo and to whatever is placed in front of them [8]. An always-available, fluent default answer is exactly the kind of anchor that theory predicts people will converge on.

The precise question. "Did AI homogenize writing?" is too coarse to test. We decompose it into three separable questions, and the paper's contribution is keeping them apart:

1. **Aggregate similarity**—are any two texts from the same period more alike than they used to be?
2. **Within-topic style**—holding the *subject* fixed, do people write it more similarly (the "everyone sounds like ChatGPT" claim in its strict form)?
3. **Topic composition**—is the *distribution of subjects* narrowing, so texts look more alike simply because everyone now writes about the same things?

A rise in (1) is compatible with (2), (3), or both. Most public commentary implicitly claims (2); we find the evidence points mainly to (3).

Framing decision (open)

Mark leans toward the affirmative "AI is driving real, consequential homogenization." The data supports that claim at the *topic* level and rejects it at the *style* level. The introduction can honor the affirmative reading by foregrounding the topic-level finding and the epistemic-diversity stakes, without overclaiming a style effect the data does not support.

2 Data

All corpora are public, dated to the quarter, and run through 2026Q2, giving roughly 14 quarters of post-ChatGPT observation against a long pre-2022 baseline. The ChatGPT marker throughout is 2022Q4.

2.1 Stack Exchange (26 communities)

From the June 2026 Stack Exchange data dump: 14 high-cognitive-load communities (e.g. Cross Validated, Philosophy, Economics, Law, Computer Science, Physics) and 12 low-cognitive-load communities (e.g. Seasoned Advice, Travel, Bicycles, Gardening, Role-playing Games, Board Games). The split tests whether casual writing homogenizes more than expert writing. Five "core" communities (Cross Validated, Philosophy, Economics, Seasoned Advice, Travel) carry every control; the other 21 carry the aggregate and significance tests.

2.2 External, non-Stack-Exchange corpora

To test whether any finding is an artifact of Stack Exchange's terse, structured Q&A style, we added four corpora from entirely different platforms (Table 1). arXiv mathematics serves as a low-AI control: unlike computer science, its subject matter has little to do with AI itself. All four carry the full pipeline, including topic control.

Table 1: External corpora.

Corpus	Register	Docs	Span	Unit
Hacker News	tech-forum comments	∼39k	2016Q1–2026Q2	comments
PubMed (oncology)	scientific abstracts	∼34k	2016Q1–2026Q2	abstracts
arXiv (Computer Science)	scientific abstracts	∼64k	2016Q1–2026Q2	abstracts
arXiv (Mathematics)	scientific abstracts	∼69k	2016Q1–2026Q2	abstracts

Framing decision (open)

Are the externals "robustness checks on a Stack Exchange result" or a "co-equal cross-domain contribution"? With four registers including a low-AI control, the latter framing is now defensible; legal opinions or corporate filings would strengthen it further.

3 Methodology

The design is a layered battery of controls, each ruling out one benign explanation for an apparent rise (Table 2). Read top to bottom, each layer strips away a confound the previous one left standing.

Table 2: The layered control design.

Layer	What it measures	What it rules out
Surface / lexical	type-token ratio, MTLD [6], Yule's K	vocabulary-diversity change
Length control	same metrics on first-100-token windows	answers simply got longer
LSA semantic	TF-IDF → SVD cosine, centroid variance	coarse topic drift
Contextual embedding	Sentence-BERT / MiniLM pairwise cosine [7, 9]	— (the live signal)
Topic control (4 ways)	within-topic, same-question, decomposition	topic-mix shift vs. style convergence
Significance	segmented regression + Mann–Kendall [5] + bootstrap	noise; wrong break date
Known-AI anchor	distance from human answers to an actual ChatGPT answer	"similar to each other" ≠ "similar to AI"

Two commitments deserve emphasis because they are where naive analyses most often go wrong.

Length control. Embedding similarity rises mechanically with text length. We recompute every headline metric on text truncated to the first 100 tokens. A signal that survives is not a "posts got longer" artifact.

Topic control, done four ways. To separate *style* from *subject* we (a) cluster with BERTopic [4] and measure within-topic similarity over time; (b) apply a Mann–Kendall trend test to the post-2022 within-topic series; (c) use the *same-question* gold standard, comparing only answers to the identical question—no clustering, no dropped data; and (d) decompose the aggregate cosine into within- and between-topic components and track topic diversity directly (global k-means, no dropped outliers). Agreement across four topic definitions is what makes the topic-composition conclusion robust.

Anisotropy. Raw embedding cosines are inflated because embeddings occupy a narrow cone [2]. Where it matters we report the mean-centred (anisotropy-removed) variant, which repeatedly flattens an apparent between-topic "convergence."

4 Results

4.1 The dramatic surface rise is a length artifact

Lexical-diversity metrics show a large apparent shift after 2022, but it tracks answer length (Figure 1): raw TTR falls and raw TF-IDF similarity rises as answers grow longer, while the length-controlled versions stay essentially flat. The effect never appears in already-verbose corpora such as Philosophy. Answers got *longer*, not more uniform.

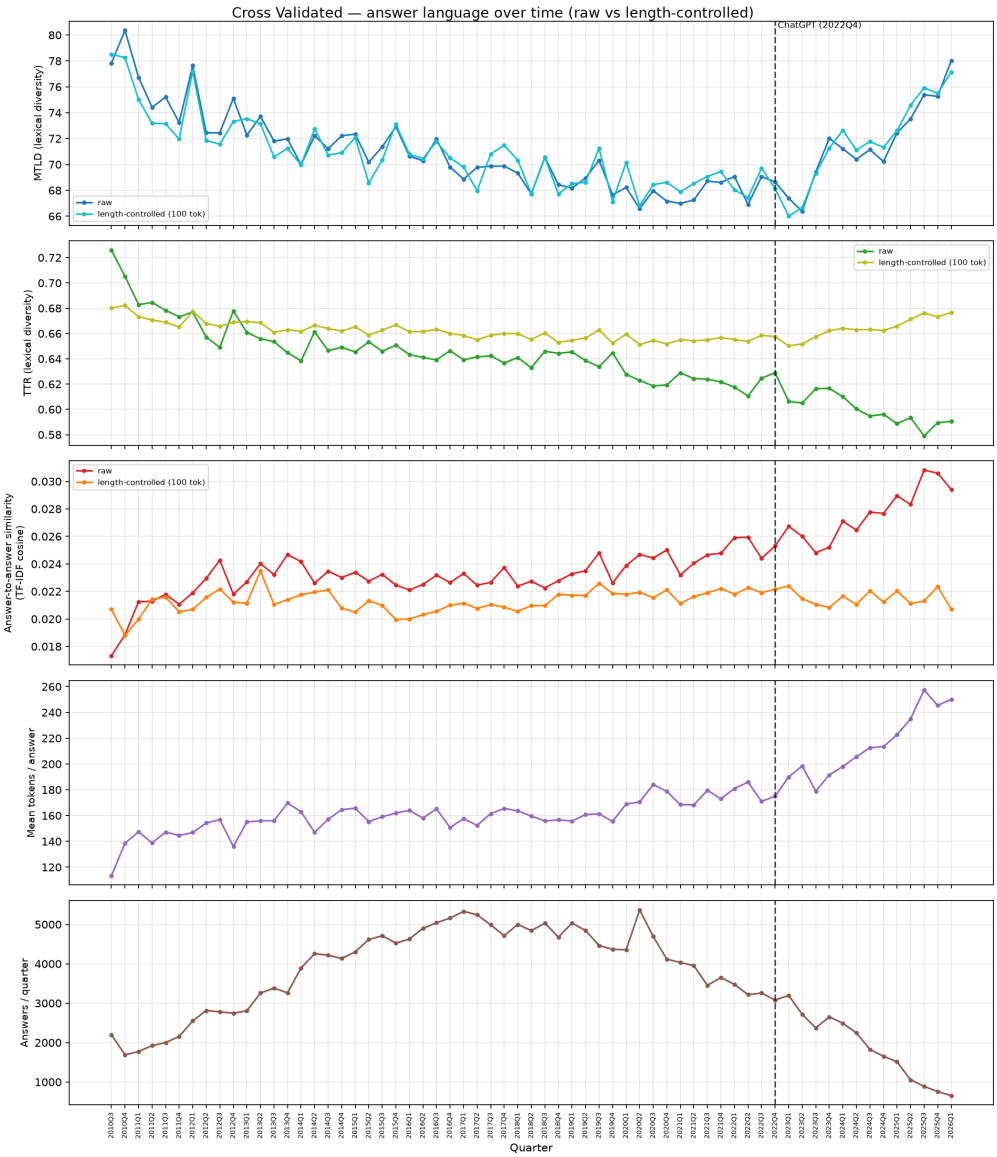


Figure 1: Cross Validated, 2010–2026. Lexical diversity (MTLD, TTR), answer-to-answer TF-IDF similarity, mean answer length, and answer volume. Raw (dark) vs. length-controlled first-100-token (light) metrics: the raw TTR decline and similarity rise follow the growth in answer length and largely vanish under length control. Dashed line: ChatGPT release (2022Q4).

4.2 A smaller contextual-embedding rise is real and survives length control

The one signal that does not dissolve: Sentence-BERT pairwise similarity rises modestly from roughly 2023Q4 onward (Figure 2), and the rise survives length control. By interrupted-time-series slope change it is statistically significant in 18 of 26 Stack Exchange corpora (20 of 26 under length control; Figure 3), and it replicates in all four external corpora: Hacker News (slope change $+0.00101$/quarter, $p = 4.5 \times 10^{-9}$; Figure 4), PubMed ($+0.00080$, $p = 9 \times 10^{-4}$), arXiv-CS ($+0.00105$, $p = 0.0014$), and arXiv mathematics ($+0.00130$, $p = 4 \times 10^{-10}$; length-controlled $p = 0.007$).

The mathematics result is notable because it is a low-AI control: before 2022 its similarity was *falling* (-0.00057/quarter, $p \approx 10^{-34}$), and after 2022Q4 that decline reverses into a significant rise

(post-break Mann–Kendall $p = 0.0013$). The reversal begins only around 2024, and the recent level is still below 2016.

Two honesty notes: the effect is broad but modest (significant partly because N is large; for Cross Validated the recent level is still below its own 2010–11 values), and PubMed and arXiv show a pre-2022 upward trend—for them the change is an acceleration, not a clean break.

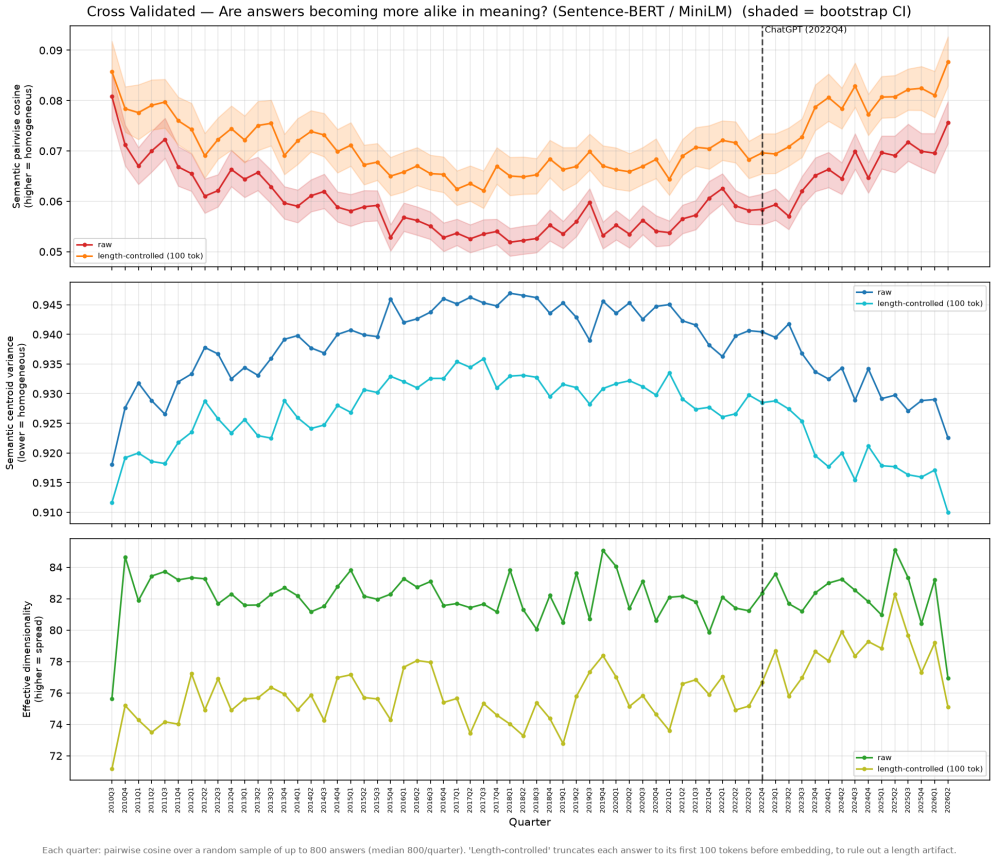


Figure 2: Cross Validated: contextual-embedding (MiniLM) similarity metrics per quarter with 95% bootstrap bands, raw and length-controlled.

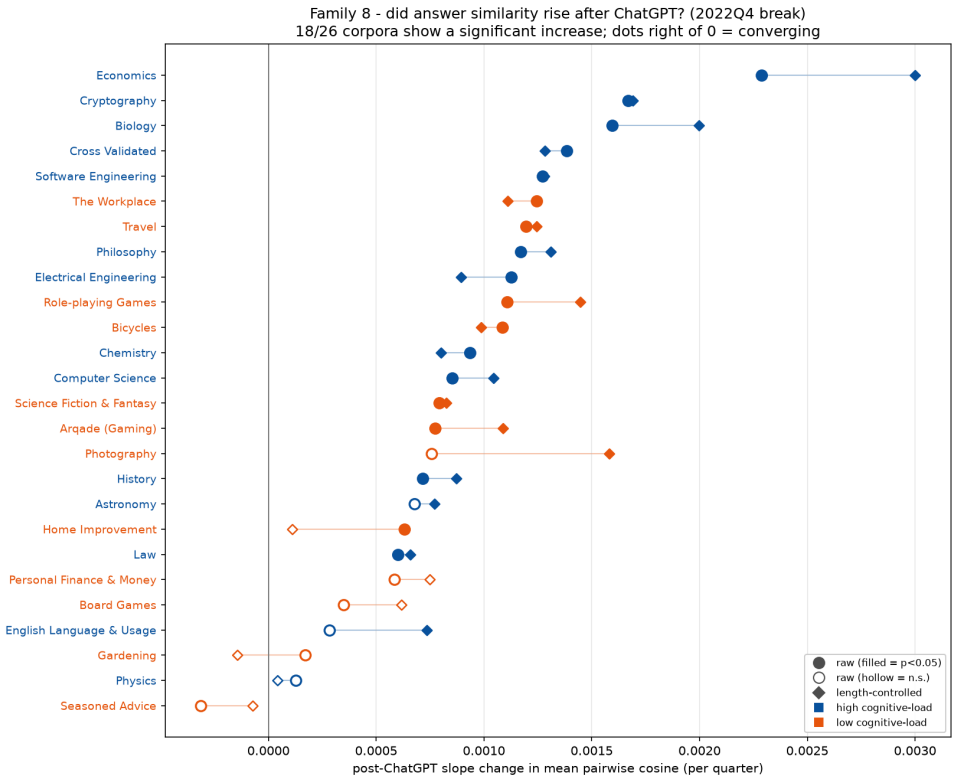


Figure 3: Post-ChatGPT slope change in mean pairwise cosine for all 26 Stack Exchange corpora (filled = $p < 0.05$; diamonds = length-controlled; blue = high, orange = low cognitive load). 18 of 26 show a significant increase.

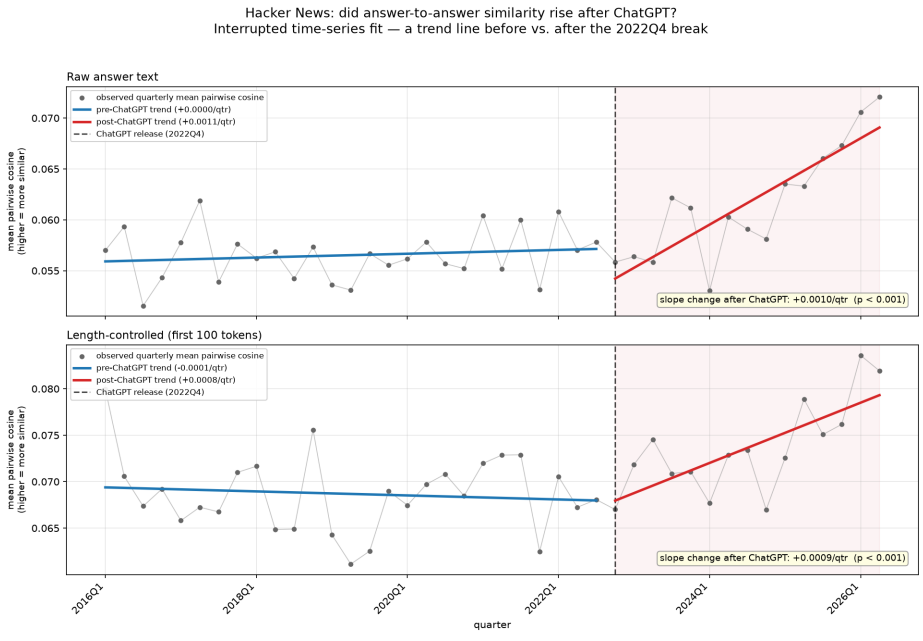


Figure 4: Hacker News: interrupted time series around 2022Q4—the cleanest external case (flat before, rising after).

4.3 The residual rise is largely topic composition, not within-topic style

This is the central result. Once the subject is held fixed, the rise mostly disappears.

Stack Exchange. Within-topic and same-question similarity is flat in 4 of 5 core corpora (Figure 5). The decomposition locates the rise in *between-topic* pairs while within-topic similarity stays flat, and shows topic diversity shrinking and the topic mix drifting after 2022 (Figure 6). Philosophy is the lone within-question exception.

External corpora. Under BERTopic—the same method as the Stack Exchange headline—all four externals show within-topic similarity flat while the aggregate rises (Figure 7). Hacker News: overall post-break Mann–Kendall $p = 1.2 \times 10^{-4}$, within-topic $p = 0.44$; PubMed: overall $p = 8 \times 10^{-4}$, within-topic $p = 0.38$; arXiv-CS: overall $p = 5 \times 10^{-4}$, within-topic $p = 1.0$; arXiv mathematics: overall $p = 0.011$, within-topic $p = 0.50$. In mathematics the decomposition also shows the effective number of topics, which had been growing before 2022, turning to shrink after it. A k-means decomposition initially flagged Hacker News as a within-topic exception ($p = 0.004$), but that did not survive a second clustering method—within-topic tests are method-sensitive. Every external also shows emergent LLM-era topics after 2022Q4 (e.g. `qa_answers_prompt`, `captioning_mllms`, `segmentation_radiomics_dl`).

The reading: the rise is real, but it is mostly people converging on *what* to write about, not on *how* to write it.

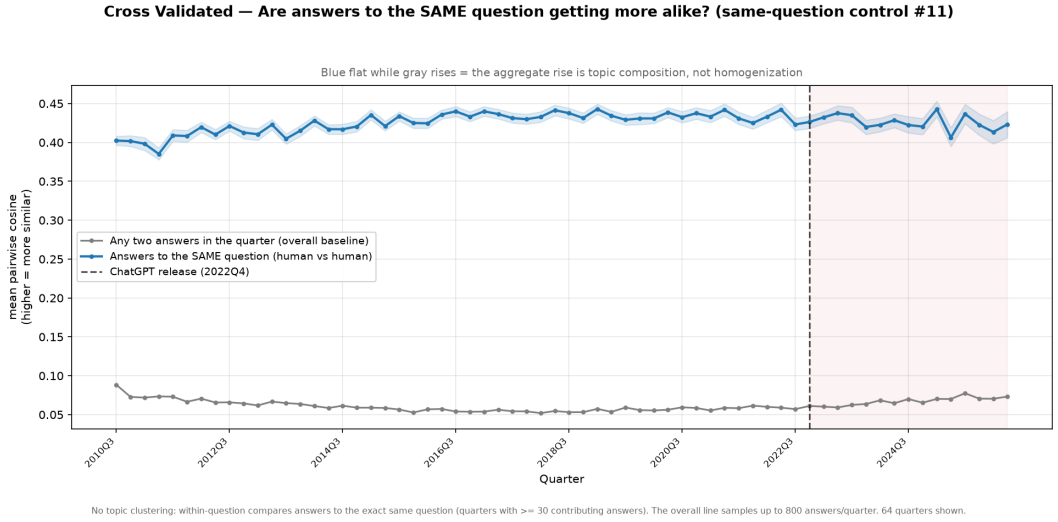


Figure 5: Same-question control (Cross Validated). Similarity between answers to the *same* question (blue) is flat across the ChatGPT break, while similarity between any two answers in the quarter (gray) rises: the aggregate rise is topic composition.

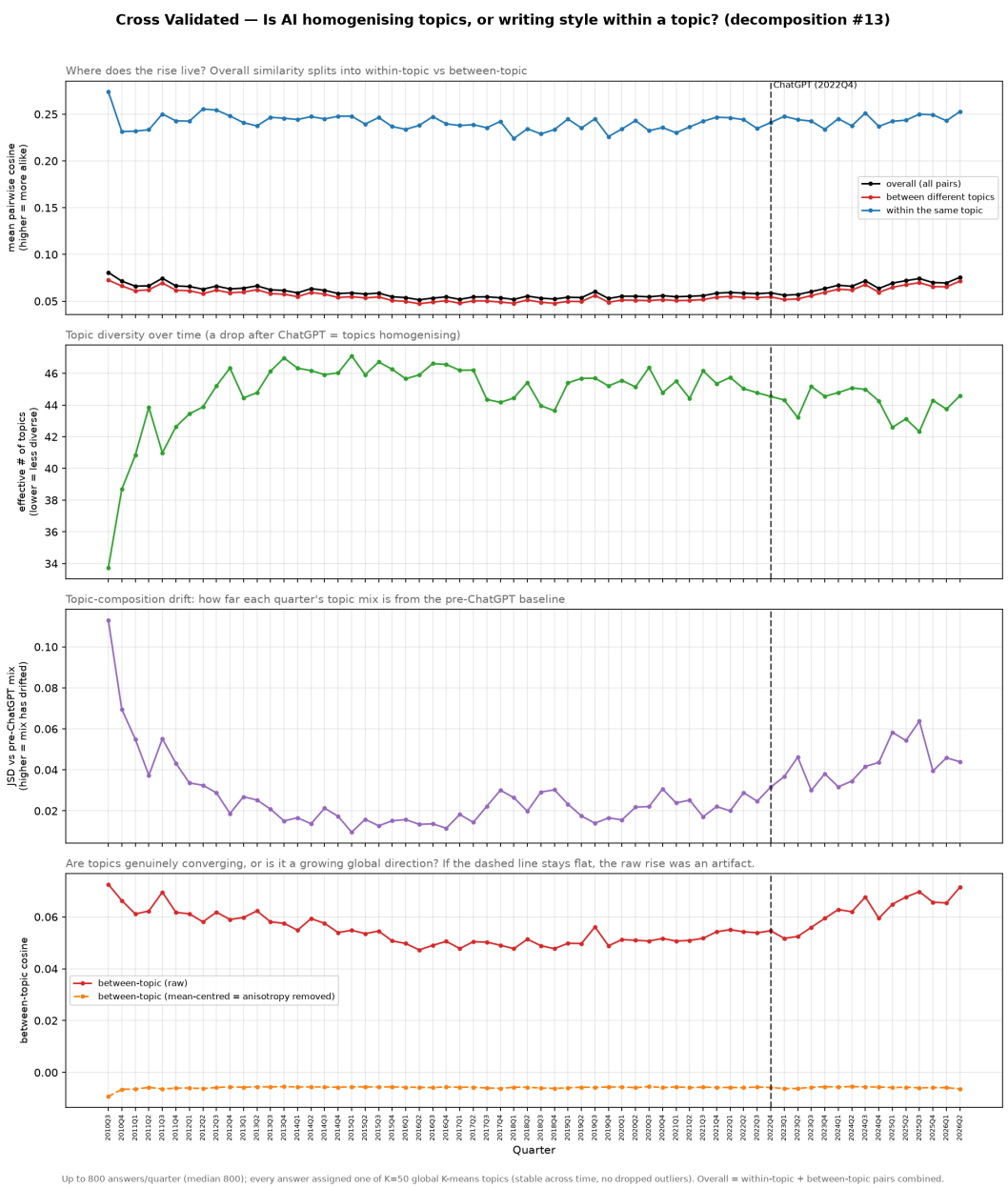


Figure 6: Within- vs. between-topic decomposition (Cross Validated; global k-means, $K = 50$). The rise lives in between-topic pairs; effective number of topics falls and topic-mix divergence from the pre-2022 baseline grows. The mean-centred between-topic line is flat (anisotropy).

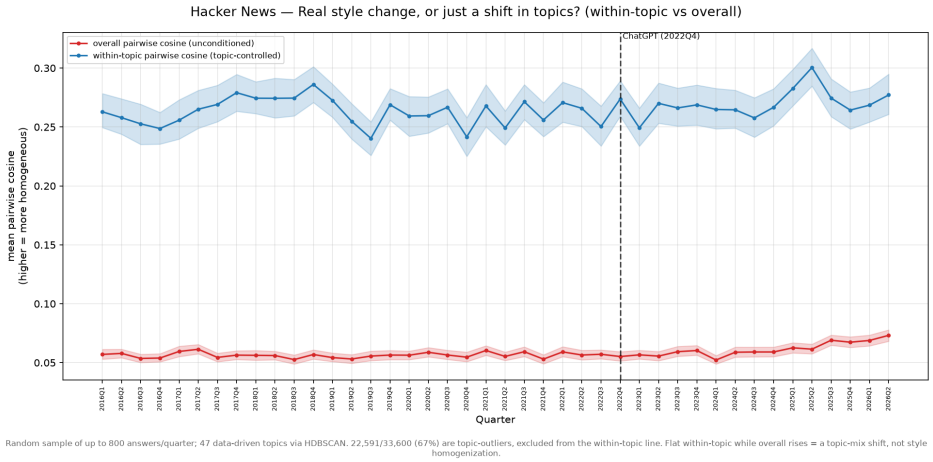


Figure 7: Hacker News: overall vs. within-topic (BERTopic) similarity. Within-topic flat while overall rises—the same topic-composition signature as Stack Exchange.

4.4 No drift toward actual AI output; cognitive load does not explain the rise

The most literal test—do human answers move toward an actual ChatGPT answer to the same question?—is flat in 4 of 5 corpora, with Economics the lone and noisiest exception (Figure 8). The rise is also independent of cognitive load: high- and low-load communities track each other (Figure 9). The two lone exceptions (Philosophy in the same-question test, Economics in the anchor test) fall on different sites and so read as noise, not corroboration.

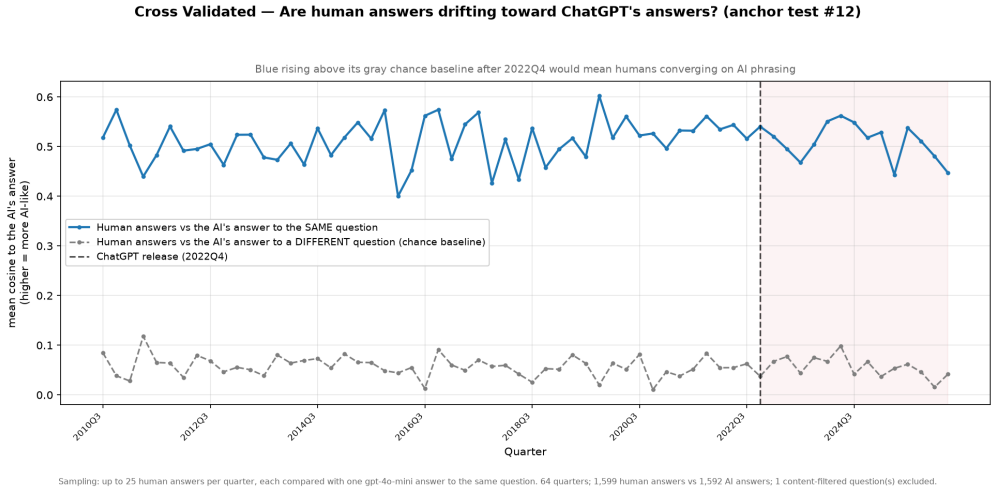


Figure 8: Known-AI anchor (Cross Validated): cosine similarity between human answers and a gpt-4o-mini answer to the same question, per quarter. No drift toward the AI answer.

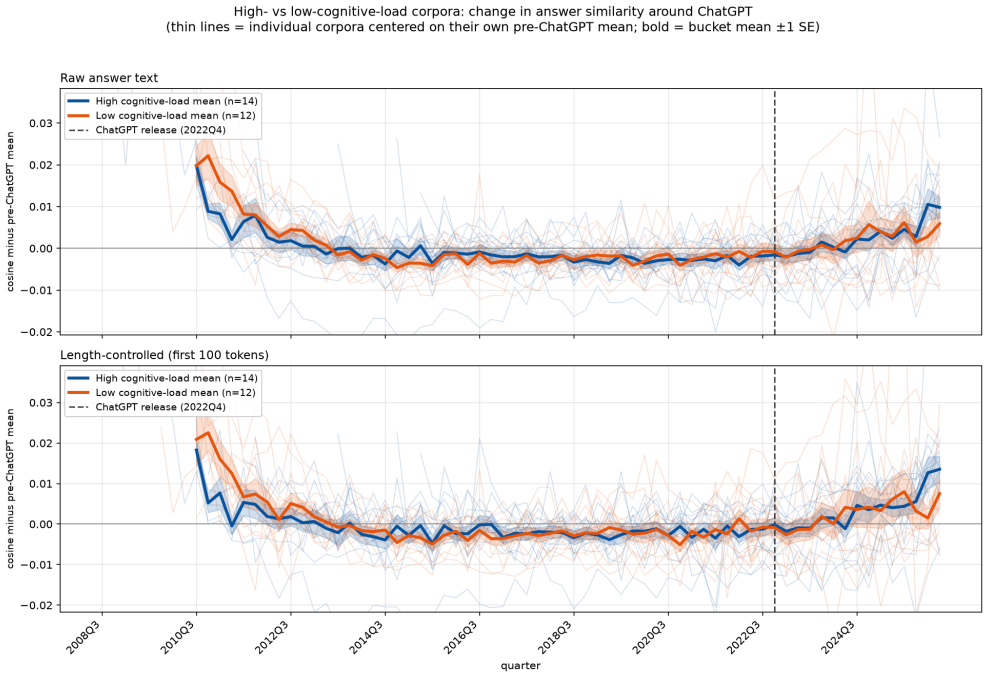


Figure 9: High- vs. low-cognitive-load communities (bucket mean ±1 SE; individual sites faint). The two tracks overlap.

4.5 Summary of evidence

Table 3: Summary of evidence.

Claim	Verdict	Backing
Surface convergence	Artifact (length)	lexical + LSA families
Aggregate embedding rise	Real, modest, broad	18–20/26 SE + 4/4 externals
Within-topic style convergence	Not supported	4/5 core SE + 4/4 externals flat
Topic-level homogenization	Supported	effective topics ↓, mix divergence ↑, emergent LLM topics
Drift toward AI output	Not supported	anchor test flat in 4/5
Cognitive-load split	None	high/low tracks overlap

5 Real-World Consequences

The finding is often heard as reassuring ("so writing *didn't* homogenize"). We argue the opposite is closer to the truth: the kind of homogenization we *did* find—topic-level—is the more consequential one.

- **Narrowing of the idea space, not just the word space.** A flat within-topic style but a shrinking, drifting topic mix means the change is in *which questions get asked and answered at all*. Fewer effective topics is a direct measure of a narrower public conversation.
- **The scientific record inherits a model's priors.** Emergent post-2022 topic clusters in arXiv and PubMed show the research frontier reorganizing around AI-adjacent themes.

Whether that is genuine progress or a bandwagon is exactly what a durable record should let us audit.

- **Search returns more pages that say less.** If top results increasingly share a topic profile, the marginal value of reading result #5 after #1–4 falls.
- **A feedback loop.** Models are trained on the web; the web is increasingly written with model help; topic-level convergence today biases the training data of tomorrow—a slow ratchet rather than a cliff.
- **Where it is *not* alarming (yet).** We find no evidence that individual voice, within a subject, is collapsing, and no drift toward literal AI phrasing. The message is targeted, not apocalyptic: *watch the topic mix, not the prose.*

Framing decision (open)

How strong a claim should this section make? From most to least cautious: (a) "a subtle but structurally important narrowing worth monitoring"; (b) "an early-warning sign of epistemic concentration"; (c) "evidence of a self-reinforcing homogenization loop."

6 Monitoring Going Forward: A Live Search-Result Tracker

The retrospective study measures the past. To measure the present, we built an open tracker that, every week, collects Google's top results for a fixed panel of 45 informational questions across five high-stakes domains (health, finance, law, science, technology), fetches each result's full page, and scores how much the pages repeat one another with the Vendi score [3]—the effective number of distinct pages—normalised by the number of pages analysed. Video results and pages that cannot be fetched are excluded. The data and dashboard are public and versioned. The series began in October 2026; with week-to-week movement still comparable to between-domain differences, trends should be read only after several months of accumulation.

7 Criticisms and Limitations

- **Significance $\neq$ magnitude.** Large N makes small slopes significant; several effects are small in absolute terms.
- **Timing $\neq$ causation.** The clearest bends lag ChatGPT by about a year and lack a clean 2022Q4 step in several corpora; PubMed and arXiv-CS were already trending up before 2022, and the arXiv-mathematics reversal only begins around 2024.
- **Topic-control fragility.** Within-topic tests depend on the topic definition: BERTopic discards 50–67% of documents as outliers, and k-means and BERTopic disagreed on Hacker News. We rely on the agreement of four methods, not any one.
- **Anisotropy.** Raw between-topic "convergence" is largely embedding geometry; the mean-centred variant flattens it.
- **Single encoder.** All embedding results use one model family (MiniLM); a second encoder is pending.

- **Register bias.** Stack Exchange answers are atypically structured; the externals mitigate but do not eliminate this.
- **Lone exceptions.** Philosophy (same-question) and Economics (anchor) each show an effect on exactly one test, on different sites; reported, not suppressed.

8 Conclusion

The popular claim that generative AI made everyone write the same way is not supported by a layered, multi-corpus, topic-controlled analysis. The large surface convergence is a length artifact; the smaller, real, length-robust embedding rise is mostly a change in *what* people write about, not *how*. What we do find is topic-level homogenization: a measurable narrowing and drift of the topic mix, and the emergence of a shared, AI-shaped set of themes across forums, science, and technical writing alike. That is a subtler but arguably more consequential effect for epistemic diversity than stylistic sameness—and, because it is measurable, it is worth monitoring in the open. The honest headline: *not broad style homogenization, but a real and structurally important topical convergence—worth watching, not yet worth alarm.*

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